

MEMORANDUM

November 14, 2025

TO: Dr. Arthur Erdman

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SUBJECT: Progress Report 2 - Analytical Synthesis and Final Constraints

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EXECUTIVE SUMMARY

This memorandum summarizes progress on the miniature tabletop greenhouse mechanism design project since Progress Report 1. We used two different methods to find the best mechanism dimensions: a MATLAB rigid solution generator and a Python optimization program. The rigid solution generator was used for Watt I mechanisms, while the Python optimization program was used for both Watt I and Stephenson I mechanisms. The current candidate solutions presented in this report are from the Python optimization framework. The MATLAB solution found a separate Watt I solution, but it has unacceptably high joint forces and is not included among the current candidates. We have clearly defined all final constraints and verified that our solutions meet them, providing a solid foundation for detailed design and prototype development.

TYPE SYNTHESIS AND LAYOUT

Building on the type synthesis completed in Progress Report 1, this section summarizes the key findings and topology selection process.

Type Synthesis Process

Following Erdman and Sandor (1990) methodology [1], we used Gruebler's equation $F = 3(n - 1) - 2f_1 - f_2$ to analyze planar mechanisms. Our initial analysis of four-bar mechanisms ($n = 4$, $f_1 = 4$) showed that no four-bar solution could fit within the space constraints. This led us to explore six-bar mechanisms ($n = 6$, $f_1 = 7$).

Four-Bar Analysis

Our initial attempts showed that four-bar linkages cannot meet both the space requirements and the door motion needs. The greenhouse is too small, and the ground pivot locations are too restricted, so no four-bar solution would work.

Six-Bar Mechanism Development

For our six-bar mechanism with one degree of freedom, we determined that we need 2 ternary links (links with three connection points) and 4 binary links (links with two connection points).

We systematically examined all possible ways to connect these links together. After eliminating configurations that wouldn't work, we found two valid options: the Watt chain, where two ternary links share a connection point, and the Stephenson chain, where the ternary links are separated by binary links.

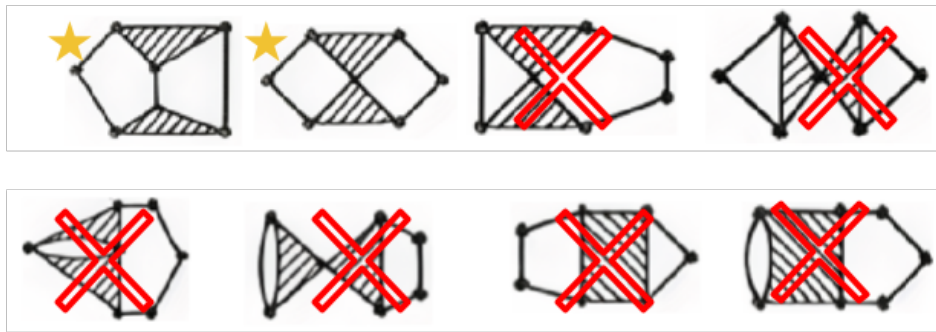


Figure 1: Six-bar mechanism topology options

Topology Selection

We evaluated five different mechanism configurations: Watt I, Watt II, Stephenson I, Stephenson II, and Stephenson III. The key difference was where the ground pivots could be placed. Watt II, Stephenson II, and Stephenson III need three ground pivots, but the greenhouse base is too small to fit three pivots. This left us with Watt I and Stephenson I, which only need two ground pivots.

Solution Selection

We selected both Watt I and Stephenson I for further development, with different team members working on different solution methods. This approach allowed us to explore more design options and compare which solution worked better.

The Watt I configuration provides:

- Two accessible ground pivots (15-20 cm separation)
- Compact dual-loop structure with 8-20 cm link lengths
- Single-degree-of-freedom motion with one input

The Stephenson I configuration offers:

- Two accessible ground pivots with greater separation potential

- Stable dual-loop structure with separated ternary links
- Single-degree-of-freedom motion with one input

ANALYTICAL SYNTHESIS

After completing the type synthesis in Progress Report 1, we developed two different methods to find the exact dimensions for the Watt I and Stephenson I six-bar mechanisms. The first method uses traditional analytical synthesis (a rigid solution generator based on Burmester theory). The second method uses a Python optimization program with differential evolution. These methods provide flexibility in meeting all the design constraints simultaneously.

Synthesis Methodology

We used two different methods to find mechanism dimensions:

1. **Rigid Solution Generator:** A MATLAB-based approach using four precision position (4PP) analytical synthesis methods based on Burmester theory, with solution rectification to eliminate branch and circuit defects.
2. **Python Optimization Framework:** A differential evolution approach that explores the design space through population-based optimization.

The Python optimization framework was used to find solutions for both Watt I and Stephenson I mechanisms. Due to time constraints, the rigid solution generator was only used to find solutions for Watt I. This approach allowed us to explore more design options and verify our results.

Rigid Solution Generator Method

The rigid solution generator uses traditional analytical synthesis methods based on Burmester theory to find mechanism dimensions. This MATLAB-based approach was developed specifically for Watt I mechanisms and uses four precision position (4PP) synthesis to break the six-bar mechanism into smaller components (triads and dyads).

Development Process

Two approaches were attempted during development:

Initial Approach - MultiStart Optimization: The first attempt used MATLAB's built-in MultiStart optimization algorithm with an objective function that solved vector loop equations, minimized the distance from the coupler point to precision positions, and optimized for maximum joint forces along with constraints on pivot and link locations. Despite many iterations with different constraint weightings and precision position adjustments, this approach did not yield useful mechanisms. The primary limitation was an inability to perform solution rectification, resulting in almost all solutions having branch and circuit defects.

Second Approach - 4PP Ground Pivot Specification: The successful approach uses 4PP ground pivot specification to generate the two triads of the Watt I mechanism, followed by 4PP synthesis to generate the connecting dyad. Compatibility linkages are used to generate Burmester curves, which represent continuous sets of valid mechanism parameters that satisfy the precision

position requirements. Solution rectification is then applied to the Burmester curves to eliminate solutions with branch and circuit defects, ensuring the mechanism follows a single continuous path. This methodology follows the approach described by Mirth [2] for solution rectification in four position synthesis of six-bar linkages.

Results and Current Status

A solution has been found that meets all geometric and motion constraints. However, the joint forces are currently unacceptably high. These forces were calculated based on an assumed spring placement for mechanism balancing. The spring location assumption may not be optimal, and future work will explore alternative spring placements or multiple springs to reduce joint forces to acceptable levels.

The mechanism does have a minor branch or circuit defect at its final precision position, but this is acceptable because the coupler curve for a single geometric inversion passes very near this position, so the mechanism will perform adequately. There is also a small hook shape in the coupler curve near the top front corner of the enclosure frame, which is less than ideal and will be addressed in future refinement work.

Python Optimization Framework

The Python optimization program provides a different approach that works well when we need to satisfy many constraints at once. It uses a differential evolution algorithm to find the best solution by trying many different designs and gradually improving them. This is especially useful when our constraints are not rigidly defined, but rather are a range of values.

Kinematic Solver

We created a motion calculator for both Watt I and Stephenson I mechanisms that calculates joint positions throughout the mechanism's motion. The solver uses Euler's formula for link endpoint calculations, circle-circle intersection methods for solving dyad positions, and triangle solution methods for ternary link configurations. These calculations are essential to determine if a mechanism design can actually move through the required path.

Objective Function

The optimization program minimizes an error function that balances how accurately the door follows its path with how well the mechanism meets all constraints. The function includes terms for start and end position accuracy, with large penalty terms for constraint violations such as forbidden zone entries, workspace boundary violations, and path continuity problems. This approach ensures that solutions not only follow the desired path but also satisfy all geometric and motion constraints.

Optimization Algorithm

We employed SciPy's differential evolution global optimization algorithm, which works well for complex optimization problems like mechanism design. The algorithm explores the design space by maintaining a population of candidate solutions and gradually improving them through mutation and crossover operations. This method is particularly effective for finding solutions when the design space has many local minima, as is common in mechanism synthesis problems.

Python Optimization Solutions

The following tables present the optimized mechanism parameters found by the Python optimization framework for both Stephenson I and Watt I topologies:

Parameter	Value
Ground Pivot A	(0.878, 0.139)
Ground Pivot G	(0.399, 0.758)
Ternary Link 2 (ABC)	
L_{AB}	0.471
L_{AC}	0.584
L_{BC}	0.476
Link 3 (BD)	0.629
Link 4 (CE)	0.523
Ternary Link 6 (EFG)	
L_{EF}	0.668
L_{FG}	0.578
L_{EG}	0.272
Link 5 (DF)	0.821
Input angle start	101.5°
Input angle end	143.8°
Rotation range	42.3°

Stephenson I solution

Parameter	Value
Ground Pivot A	(0.974, 0.542)
Ground Pivot G	(0.528, 0.647)
Link 1 (AB)	0.288
Ternary Link 2 (BCE)	
L_{BC}	0.480
L_{CE}	0.246
L_{EB}	0.521
Link 3 (CD)	0.940
Ternary Link 4 (EFG)	
L_{EF}	0.597
L_{FG}	0.458
L_{GE}	0.256
Link 5 (DF)	0.602
Input angle start	−159.4°
Input angle end	−36.0°
Rotation range	123.4°

Watt I solution

Table 1: Python optimization results for Stephenson I and Watt I mechanisms (normalized units)

Results and Analysis

The Python optimization process found several working solutions for both mechanism types. We identified four candidate mechanisms from the Python optimization and ranked them by how well they performed. Note that these solutions are distinct from the MATLAB rigid solution generator results, which found a separate Watt I solution with different characteristics.

Performance Metrics (Python Optimization Solutions):

- Start position accuracy: < 0.01 units from target
- End position accuracy: < 0.1 units from target
- Continuous path: Verified across full motion range
- Forbidden zone clearance: Maintained throughout motion
- All joints remain in valid workspace

The Stephenson I mechanism from the Python optimization performed best because it has better separation between ground pivots (0.78 units based on the optimized solution) compared to

the Watt I solution (0.46 units), and better force transmission angles throughout the motion. This configuration has two ternary links (ABC connected to ground at A, and EFG connected to ground at G) joined by three binary links, creating a stable two-loop structure. Three Watt I configurations were also found by the Python optimization and ranked second through fourth place. While all Watt I solutions from Python optimization met the main path requirements, they had ground pivots closer together and needed larger input rotations (123.4° for the solution shown), which could make it harder to design the counterbalance system. The MATLAB rigid solution generator found a separate Watt I solution with different parameters, but that solution has unacceptably high joint forces and is not included in the current candidate solutions.

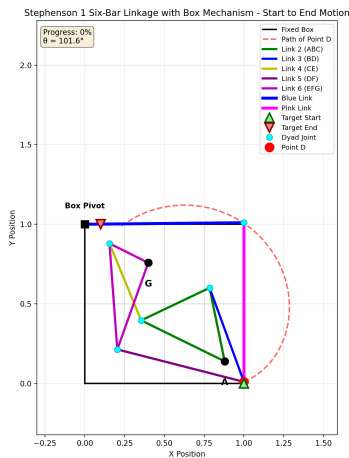
Visualization and Verification

We verified each mechanism design by:

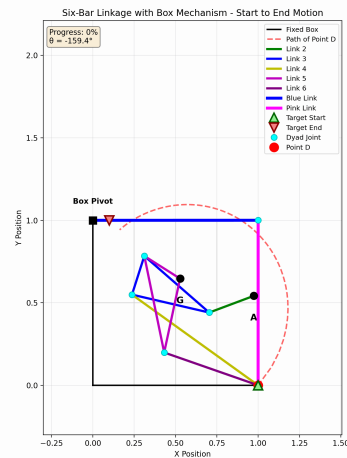
1. **Path plots** showing the door's path from closed to open position
2. **Range visualizations** showing the full range of motion
3. **Box diagrams** showing how the mechanism fits inside the greenhouse
4. **Animations** showing smooth continuous motion

All visualizations confirm that the mechanisms successfully:

- Start with Point D at (1.0, 0.0) - the closed position at lower right corner of enclosure
- Move Point D to (0.1, 1.0) - the open position near upper left
- Avoid the forbidden circular zone centered at (0, 2) with radius 1
- Keep all links outside the box walls during the opening motion
- Maintain geometric feasibility throughout the motion range



Stephenson I mechanism



Watt I mechanism

Figure 2: Stephenson I and Watt I chain configurations in motion

FINAL CONSTRAINTS

The following constraints define what the final mechanism must do.

Motion Limits

The mechanism must move the door attachment point (Point D) through a specific path:

- **Start position:** Point D at $(1.0, 0.0) \pm 0.01$ units (normalized coordinates) - represents the closed configuration with door panel at the lower-right corner of the enclosure
- **End position:** Point D at $(0.1, 1.0) \pm 0.1$ units - represents the open configuration with door panel folded upward
- **Motion continuity:** The mechanism must follow a smooth, continuous path between these positions with no path branching or mechanism defects
- **Input rotation range:** The mechanism must be actuated through the required input angle range (approximately 42° for Stephenson I, 123° for Watt I based on Python optimization results)

Workspace Boundaries and Ground Pivot Regions

- **Acceptable ground pivot regions:** Both ground pivots (A and G) must be located within the normalized box $[0, 1] \times [0, 1]$, corresponding to the greenhouse base footprint (approximately $46 \text{ cm} \times 46 \text{ cm}$)
- **Closed position workspace:** All joints must be within the $[0, 1] \times [0, 1]$ box when the mechanism is in the closed position, ensuring the mechanism fits completely inside the greenhouse when closed
- **Forbidden zone:** The workspace boundary is defined by a forbidden circular zone centered at $(0, 2)$ with radius 1 (normalized units), representing the wall behind the box. Point D must maintain at least 1.0 units distance from this center throughout the motion to avoid interference
- **Motion boundaries:** No joint may have negative x or y coordinates during motion (left wall and floor constraints)

Force and Load Restrictions

- **Panel weight support:** The mechanism must support approximately 2.3 kg (5 lb) combined panel weight
- **Smooth motion:** The mechanism must provide smooth velocity and acceleration profiles to prevent binding or vibration

Practical Limits

- **Size constraints:** Compact tabletop dimensions of approximately 46 cm × 46 cm × 61 cm
- **Materials and manufacturing:** Link dimensions must be suitable for laser cutting or machining, with joint spacing allowing standard bearing installation
- **Position holding:** The mechanism must remain stable in any intermediate position between fully open and fully closed, allowing the front and top panels to be positioned at any point and remain in place

How Constraints Affect Final Design Decisions

These constraints directly influenced our mechanism selection and optimization process. The workspace boundary constraint (forbidden zone) and ground pivot location requirements eliminated four-bar mechanisms and constrained our six-bar solutions to Watt I and Stephenson I topologies with only two ground pivots. The motion limits requiring precise start and end positions led us to use analytical synthesis methods (Burmester theory) and optimization algorithms that could simultaneously satisfy multiple precision positions.

The Stephenson I solution from the Python optimization was selected as the best-performing mechanism because it provides better ground pivot separation compared to the Watt I solutions, which improves force transmission and allows for more flexible mounting options. The Stephenson I also requires a smaller input rotation range (42.3°) compared to the Watt I solution shown (123.4°), making actuation system design more practical. The optimized Stephenson I solution successfully satisfies all constraints, providing a solid foundation for detailed design and prototype development.

CURRENT STATUS AND NEXT STEPS

We have successfully completed the type synthesis and analytical synthesis phases of the project. Watt I mechanisms were analyzed using both the rigid solution generator and Python optimization methods, while Stephenson I mechanisms were analyzed using the Python optimization method. The current candidate solutions presented in this report are from the Python optimization framework. The MATLAB rigid solution generator found a separate Watt I solution, but it has unacceptably high joint forces and is not included among the current candidate solutions. Working solutions have been identified for both mechanism types that satisfy the design constraints. The Stephenson I topology from Python optimization has shown superior performance in meeting the path generation and workspace requirements.

The immediate next steps are to:

- Decide on a final solution between the candidate mechanisms
- Investigate and implement a spring system for balancing the mechanism
- Build a scaled prototype once the spring system design is complete

These steps will validate the theoretical synthesis results and provide practical insights for the final design phase.

TEAM MEMBER CONTRIBUTIONS

- **Sebastian Hondl:** Python optimization algorithm
- **Will McConnell:** MATLAB solution generator
- **Peyton Lettau:** Reports, presentations, and solution assistance
- **John Robinson:** Solution assistance

CONCLUSION

We have successfully found working mechanism dimensions for both Watt I and Stephenson I six-bar mechanisms using the Python optimization framework. The MATLAB rigid solution generator also found a Watt I solution, but it has unacceptably high joint forces and is not included among the current candidate solutions. Using analytical synthesis and optimization methods, we confirmed that six-bar mechanisms can meet all the space and motion requirements for the miniature greenhouse. We have clearly defined and verified all final constraints. This work provides a solid foundation for detailed design, force analysis, and prototype development in the final project phase.

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References

- [1] A. G. Erdman and G. N. Sandor, *Mechanism Design: Analysis and Synthesis*, vol. 1. Englewood Cliffs, NJ: Prentice Hall, 2nd ed., 1990.
- [2] J. A. Mirth, *Solution Rectification for the Multiple Circuit and Transmission Angle Problems in Four Position Synthesis of Six-Bar Linkages*. PhD thesis, ProQuest Dissertations & Theses, 1990.